# Willian Vieira PhD

Data engineer and scientific modelling specialist with a PhD in quantitative ecology. I build data workflows, ETL pipelines, and reproducible systems that turn messy, decision-critical data into validated, traceable outputs people can trust.

## DATA SCIENCE & ENGINEER EXPERIENCE

2025 | 2024 (1 yr 9 m)

### Data Analyst & developer

Habitat, Montreal, Canada

Built production-minded Python/R analytical pipelines that turned messy real-world project data into reusable, validated model inputs and decision-ready outputs | Developed metadata-driven ETL and provenance workflows so datasets, definitions, analyses, and handoffs remained traceable, reproducible, and inspectable | Led the transition to a Unix-based production environment using Docker, CI/CD, automated testing, documentation, and collaborative development practices | Designed cloud/Azure-oriented data infrastructure that made heterogeneous analytical datasets easier to retrieve, validate, and reuse

2024 | 2017

### PhD Research

Integrative Ecology Lab, Sherbrooke, Canada

Developed Bayesian hierarchical models, simulation workflows, and historical-data evaluation routines to forecast forest dynamics from noisy, biased, incomplete real-world data, with explicit uncertainty propagation and model comparison | Ran thousands of simulations on HPC infrastructure () to test algorithmic assumptions, compare model behavior, and scale analyses beyond local compute | Built reproducible pipelines for continental-scale geospatial and environmental data, harmonizing climate, satellite, and field-inventory datasets into modeling-ready inputs | Developed open-source software libraries (, ) for demographic modelling with version control, automated testing, documentation, and reusable workflows | Published a technical methods book documenting the full computational pipeline, along with one peer-reviewed publication and two preprints (, )

2022 | 2020 (part-time)

### Biostatistician

Environment and Climate Change Canada - Quebec, Canada

Developed a cost-aware probability sampling protocol to improve the spatial representativeness of boreal bird surveys in Quebec | Led R&D on spatial bias-correction methods for large-scale ecological monitoring data, designing an approach later adopted by other provinces | Engineered a fully automated and reproducible data pipeline with version-controlled workflows, automated documentation, and stakeholder-ready analytical outputs

## EDUCATION

2024 | 2017

### PhD, Ecology

Université de Sherbrooke - Sherbrooke, Canada

*How climate, competition, and forest management shape the limits of tree species distributions: from individuals to metapopulations*

2016 | 2015

### Masters 2, Agroecology and Resource Management

Bordeaux Sciences Agro, Bordeaux, France

*Modelling the dispersion of weed species in agricultural landscapes*

2015 | 2010

### BSc in Agronomy

Universidade Federal de Santa Catarina, Florianópolis, Brazil

## CONTACT


(263) 381-3636

w.vieiraw@gmail.com

Linkedin/WillVieira

GitHub/WillVieira

Publications


## SKILLS

**Data Engineering & Analytics**: Python · SQL · R · ETL/ELT · DuckDB · Data modelling · Transformations · Data quality

**Scientific Modelling & Statistics**: Bayesian inference · Hierarchical models · Simulation · Optimization · Model evaluation · Uncertainty

**Production Data Systems**: Reproducible pipelines · Workflow automation · APIs · Docker · Azure · AWS · Metadata · Provenance

**DevOps & Collaboration**: Git · CI/CD · Unix · Unit testing · Automated validation · Documentation · Stakeholder communication

## OUTREACH

· Developed and maintained the QCBS R Workshop Series, reaching nearly one thousand graduate students · Taught programming, statistics, SQL, and reproducible workflows to 250+ students across biology, engineering, and graduate programs · Translated stakeholder needs into technical analyses, documented pipelines, and clear analytical outputs

## LANGUAGES

Portuguese · Native

French · Full Professional

English · Full Professional

*Source · HTML · PDF*

*Updated 2026-06-09*